

Below is a comprehensive development blueprint and functional synopsis of the physical drive intake, replication, and duplication management system. This technical overview is structured specifically to serve as a high-density reference that you can supply to an AI coding assistant to reconstruct, extend, or migrate this application with maximum efficiency.

Application Development Blueprint: Media Duplication & Intake System

1. System Overview & Core Concept

The application is a full-stack, responsive, and secure platform designed to manage the physical intake, specification scanning, duplication tracking, and dispatching of hard disk drives (HDDs) and solid-state drives (SSDs). It is divided into four distinct operational interfaces:

- **Master Portal (Admin Portal):** The workstation where admins can manually correct data, review or reprint claim tickets and asset tags.
- **Volunteer Portal (Intake and Return Portal):** The entry point where volunteers register new drives, verify specifications, assign sources to drives, print asset tags and tickets, and lookup drive status. Has access to the Kiosk Terminal.
- **Processing Portal(Technician Portal):** The workstation view where technicians monitor active duplication progress, manage copy-cycle phases, and update drive processing status (Pending, Duplication Active, Copy Failed, Copy Complete). Has access to the Volunteer Portal and Kiosk Terminal.
- **Kiosk Terminal (Client Interface):** A customer-facing kiosk that provides instant status checks showing the current phase of duplication (Pending, Duplication Active, Copy Failed, Copy Complete). This cannot make any data changes and does not require authentication or authorization.

2. Technical Stack & Dependencies

Frontend: React 18+ (SPA structure), Vite, Tailwind CSS, motion/react for smooth transitions.

Backend: Express (Node.js) acting as an API proxy layer, managing drive inventory, and integrating with the Gemini API for live OCR vision extraction.

Key Packages:

- lucide-react: For clean, uniform technical UI iconography.
- qrcode.react (QRCodeSVG): For embedding robust vector-based QR codes on labels and claims.

Configuration Requirements:

- The backend server runs on host 0.0.0.0 and port 3000 to properly route container traffic.
- Standard TypeScript configuration (tsc --noEmit check) is integrated.

3. High-Fidelity Feature Specifications

A. Intelligent Ingest & Label Scanner (Gemini Vision OCR)

Image Upload & Parsing: Operators take a photo of a hard drive's manufacturer sticker via an attached webcam. The system routes the image through a backend endpoint (/api/disks/scan-label) leveraging Gemini Vision to extract metadata: Manufacturer/Brand, Model Spec, Serial Number Code, Capacity Size, and Spindle rotational Speed.

Unresolved Field Escalation (Mismatch Alert): If the OCR scanner returns incomplete, generic (Unknown, N/A), or wildcard (?) parameters, the system triggers a Scan Warning Overlay. The warning displays an inline checklist of missing/unresolved attributes alongside a zoomed image proof of the uploaded sticker. The operator is presented with two explicit choices:

- Manually Correct / Enter: Instantly focuses and highlights the corresponding manual registration inputs with an amber-pulsed ring to guide manual entry.
- Skip Entry of Missing Data: Bypasses the warning and allows registration with incomplete fields.

B. High-Fidelity Claims & Asset Label Generation

Each registered drive generates two separate high-contrast, black-and-white print layouts representing physical label sheets:

Physical Media Asset Tag: Formatted to be printed for affixing directly onto the drive cover. It should be sized 2 inches wide by 4 inches long. The tag features:

- The generated unique Sequence ID (Disk ID) and its associated vector QR Code.
- The source drive requested
- The disk serial number
- A photo image proof of the disk extracted during scanning.

Customer Claim Ticket: A dual-purpose thermal receipt with:

- The generated unique Sequence ID (Disk ID) and its associated vector QR Code.
- The source drive requested
- Plain-language disclaimers instructing the customer on claim processes.

C. Safety Check & Verification Protocols

Redundant Ingestion Guard: When parsing a new drive image, the system cross-references the entered Serial Number (S/N) with all active database records. If a matching serial number is found, a warning prevents duplication of the ingest action and informs the operator of the existing item's status. Manual intervention by an Admin would be required to proceed.

Phase-Locked Progress Routing: Progress statuses are locked to physical tracking pipelines:

1. Phase 01: Received / Pending
2. Phase 02: Duplication Active
3. Phase 03: Copy Complete / Dispatch Ready
4. Phase 04: Drive returned

4. Key UX & UI Design Decisions

- **Aesthetic Pairing:** Consists of Space Grotesk or clean Sans UI for displays, paired with JetBrains Mono for low-level technical parameters, serial numbers, and logs.
- **Theme Continuity:** High-contrast layout utilizing dark slate elements (bg-[#0E0E10], border-[#2A2A2E]) for admin tasks, juxtaposed against clean paper-white physical prints (bg-white, text-slate-900) for the ticket previews to simulate actual high-contrast thermal ink printing.
- **Interactivity:** Interactive buttons explicitly prompt actions (Simulate Seagate 8TB Scan, Simulate WD 6TB Scan, Simulate Mismatched Scan) to allow rapid testing of success states and error exceptions under various edge-case flows.